



Numerical simulation of thermal flow of power-law fluids using lattice Boltzmann method on non-orthogonal grids

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ABSTRACT

In this study, a lattice Boltzmann model is proposed to investigate thermal flow of power-law fluids in irregular geometries. Non-orthogonal grids on the physical plane are adopted to accurately depict curved boundaries. With respect to power-law fluid, the evolutions of particle distribution functions for velocity and temperature fields on the computational plane are derived on the basis of the generalized form of interpolation-supplemented lattice Boltzmann method and the rheological equation. For forced convection, the double-distribution-function method is used to predict the temperature field. The coupled lattice BGK model is employed to consider the conditions under which the velocity and temperature fields are strongly coupled, such as natural convection. The adiabatic boundary and symmetry boundary in body-fitted coordinates are specifically prescribed. Numerical procedure is further validated by modeling a series of cases including forced convection over a stationary heated circular cylinder and natural convection in cavities and an annulus between concentric circular cylinders, respectively. Good agreements with available data in the literature are achieved. The numerical results demonstrate that the effect of rheological and thermodynamic properties of power-law fluid on the heat transfer performance and flow field in complex geometries can be suitably captured by present thermal lattice Boltzmann method on non-orthogonal grids.

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1. Introduction

Forced and natural convections have received considerable attention due to their importance in engineering applications, such as electronic cooling systems, nuclear reactors, MEMS devices, solar energy collectors, and heat exchangers [1]. A large body of literature are focused on forced convection [2–5] and natural convection [6–11] of Newtonian fluids. However, in many practical applications such as petroleum drilling, pulp paper, food processing, geophysical system, and polymer engineering, the fluid exhibits complex non-Newtonian behavior. Considering the effect of rheological properties of non-Newtonian fluid on the hydrodynamics and thermodynamics, great efforts have been made on the studies of flow characteristics and heat transfer performances of non-Newtonian fluids.

A theoretical analysis of steady laminar natural convection heat transfer to power-law fluids was presented by Acrivos [12]. The well-established expression for the heat transfer rate to Newtonian fluid was generalized to include the non-Newtonian effects.

Pittaman et al. [13] performed an experimental investigation on laminar natural convection of shear-thinning polymer solutions from an electrically-heated vertical plate, and obtained the dimensionless relation between Nusselt, Grashof and Prandtl numbers under conditions of constant surface heat flux. Recently, the nuclear Magnetic Resonance Imaging (MRI) technique was applied to study Rayleigh-Bénard convection of shear-thinning fluids in a cylindrical cavity. Compared with the Newtonian case, it is found out that the intensity of convection is increased by using shear-thinning fluids [14]. Numerical simulation is a complementary research approach to experimental and theoretical studies. Various numerical methods were applied to study natural convection of non-Newtonian fluids in past decades. The first investigation was attributed to Ozoe and Churchill, who developed a finite difference algorithm to study natural convection of a power-law fluid in a rectangle heated from below [15]. Later, this method was applied to investigate natural convection of power-law fluids in various enclosures, such as vertical rectangular cavity [16], square cavity [17], shallow cavity [18], and inclined rectangular enclosure [19]. Barth and Carey [20] reported a benchmark solution for natural convection of a shear-thinning fluid described by Powell-Eyring model in a cubical cavity using the finite element method. The

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finite volume method was also employed to study non-Newtonian heat transfer problems [21–26]. Turan et al. [21] investigated steady two-dimensional natural convection of power-law fluids in a square enclosure with differentially heated side wall subjected to constant temperatures. A new correlation was proposed for the mean Nusselt number in response to variations of Rayleigh number, Prandtl number, and the power-law index. Khezzar et al. [22] studied the effects of inclination angle and aspect ratio on natural convection of power-law type fluids in a two-dimensional rectangular tilted enclosure. Matin et al. [23] discussed the dependence of Nusselt number on eccentricity and aspect ratio for natural convection of power-law fluids between eccentric horizontal square ducts. These studies were focused on the cases with constant temperature boundary. Turan et al. [24] analyzed the influence of constant wall heat flux on laminar natural convection of power-law fluids in a rectangular enclosure. For transition buoyant convection of power-law fluids which is beyond the scope of present work, readers can refer to the work of Kim et al. [25] and references therein.

The heat transfer performances of power-law fluids in regular enclosures have been widely studied. However, the flow domain can be a complex geometry for most practical applications. For example, the wavy surfaces are beneficial to heat transfer rate [1]. Under such circumstances, the traditional numerical simulation methods, such as finite difference, finite volume, and finite element methods, involve tedious grid generation and complicated solution process [27]. In terms of numerical approaches for fluid flow in complex geometry, it can be divided into three categories: the unstructured mesh technique, the immersed boundary method on Cartesian grids, and the boundary-fitted curvilinear method [28]. The unstructured mesh technique can have a different number of neighbors, so it has a great flexibility in adapting the local grids to the arbitrarily shaped boundaries [9]. Bharti et al. [29] numerically investigated the heat transfer from an unconfined elliptical cylinder by using the finite volume method on unstructured non-uniform grids to yield the cylinder surface. Sasmal and Chhabra [30] adopted the unstructured cells with non-uniform spacing to study laminar natural convection from a heated square cylinder immersed in power-law liquids. The immersed boundary method was firstly proposed by Peskin [31] to analyze the blood flow patterns around heart valves. In terms of irregular boundaries, a body force was introduced in the momentum equation on Cartesian grids to avoid the generation of body-fitted grids. Ren et al. [32,33] applied this method to simulate thermal flow of Newtonian fluid with Dirichlet and Neumann temperature conditions, respectively. Zhang et al. [34] developed an immersed boundary method to study forced convection of Newtonian fluid over a circular cylinder. In the third methodology, the grids in physical domain are non-orthogonal to coincide with irregular boundaries. The governing equations can be solved numerically by the finite difference method [35] and the finite volume method [36] in the computational domain with uniform grids. Thus, the grid generation and transformation of governing equations from the physical domain to the computational domain are required. Some interesting results on natural convection of power-law fluids can be found in the literature [37,38].

In recent years, the lattice Boltzmann method (LBM), which is viewed as a particle-based numerical approach at mesoscopic level, has become a powerful approach for modelling isothermal and non-isothermal fluid flows. Different from the Navier-Stokes solvers, the LBM is based on kinetic theory. This feature provides a way to include microscopic dynamics at fluid-fluid and fluid-solid interfaces. Due to particulate nature and local dynamics, the LBM has many attractive advantages such as the ability of incorporating microscopic interactions, the capability of handling complex geometry, the simplicity of programming and the parallel

algorithm [39]. As for application of the LBM in non-Newtonian fluid flows, first attempt was attributed to Aharonov and Rothman [40]. They proposed a new lattice Boltzmann model for power-law fluids in 1993. They allowed the local value of the viscosity at each lattice node to be dependent on strain rate tensor, consequently the dimensionless relaxation time at each node can be determined. Since then, the non-Newtonian LBM models for Bingham fluid [41], Herschel-Bulkley fluid [42], and viscoelastic fluid [43] have been developed successively. Particularly, Boyd et al. [44] pointed out that the strain rate tensor can be calculated locally at each lattice node in the LBM, which made this scheme efficient by avoiding the calculation of velocity derivatives. Nazari et al. [45] explored the thermal behavior of power-law fluids in a channel with a built-in porous square cylinder by using the lattice Boltzmann method. Delouei et al. [46] studied the unconfined flow and heat transfer of a power-law fluid over a heated cylinder by an immersed boundary-thermal lattice Boltzmann method. Additionally, the finite difference lattice Boltzmann method (FDLBM) was adopted to simulate natural convection of non-Newtonian molten polymer in sinusoidal heated cavity [47].

In the framework of LBM, the adaptive meshes refinement method and inter/extrapolated curved boundary method were presented to deal with curved boundaries [48–51]. Recently, Chen et al. [52] used the curved and moving boundary methods to simulate convective heat transfer of power-law fluids in an impeller stirred tank involving complex moving boundaries. All these techniques were performed on uniform grids. In addition, the interpolation-supplemented LBM (ISLBM) on non-uniform grids has been proved to be efficient in comparison with the conventional LBM for the simulation of fluid flow in complex geometries [53]. Based on irregular grids, the particles at nodes cannot move to their neighbors, so an interpolation step is introduced to determine the particle distribution functions at the lattice nodes after the streaming steps. It is noted that an analytical function is required to describe the transformation between the physical plane and the computational plane, which hampers its further application. Inspired by grid generation in body-fitted coordinates, Imamura et al. [54,55] developed the generalized form of interpolation-supplemented LBM (GILBM) and successfully applied it to simulate flow around a circular cylinder and NACA0012 airfoil, respectively. The GILBM breaks through the limitation of ISLBM and the computational efficiency is improved as well [56]. Later, Li et al. [57] developed a 2D multi-relaxation-time (MRT) lattice Boltzmann model on non-uniform mesh to study fluid flow in configurations of curved boundary. Mirzaei and Poozesh [58] also developed an interpolation lattice Boltzmann method using Joukowski transformation to link the computational plane and the physical plane, and applied it to study the 2D fluid flow around an airfoil in body-fitted coordinates.

Based on the foregoing analysis, it is evident that although the lattice Boltzmann method on non-uniform grids has been applied to hydrodynamics, very few attempts have been made to solve the energy equation, let alone the heat transfer problem involving non-Newtonian fluids. Li et al. [59] proposed a thermal LBM model on non-orthogonal grids for Newtonian fluid and validated it by simulating flow around a circular cylinder and natural convection in horizontal concentric and eccentric cylindrical annuli, respectively. The purpose of this article is to develop a thermal LBM model on non-uniform grids for heat-transfer application of power-law fluids. The present method adapts the GILBM model for the fluid motion in body-fitted coordinates. In terms of thermal flow, the double-distribution-function method proposed by Guo et al. [60] is introduced to consider the temperature evolution. The adiabatic boundary and symmetry boundary in body-fitted coordinates are derived. The accuracy of the simulation method is verified by simulating forced convection over a stationary

isothermal circular cylinder and natural convection in cavities and an annulus between concentric circular cylinders, respectively. The rest of the paper is organized as follows: In Section 2, the mathematical model for power-law fluid is introduced, followed by the thermal LBM model for power-law fluid on non-uniform grids. In Section 3, the boundary treatment is presented. Fluid flow and heat transfer simulations are given in Section 4. Finally, some conclusions are drawn in Section 5.

2. Methodology

2.1. Mathematical model for power-law fluid

In the present work, only steady two-dimensional viscous incompressible flows are considered. The governing equations including the mass, momentum and energy equations can be expressed as

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\rho \nabla \cdot (\mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} \quad (2a)$$

$$\rho \nabla \cdot (\mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} - \rho \mathbf{g} \beta (T - T_0) \quad (2b)$$

$$\mathbf{u} \nabla T = D \nabla^2 T \quad (3)$$

where ρ is the density, $\mathbf{u}$ is the fluid velocity, p is the pressure, $\mathbf{g}$ is the gravitational acceleration, T is the fluid temperature, T_0 is the reference temperature. $D = k/\rho c_p$ is the thermal diffusivity of the fluid with k and c_p being the thermal conductivity and specific heat at constant pressure, respectively. β is the coefficient of thermal expansion. $\boldsymbol{\tau}$ is the shear stress tensor. Note that Eqs. (1), (2a), and (3) are for forced convection, while Eqs. (1), (2b), and (3) are for natural convection. In Eq. (2b), the variation of density with temperature is considered using Boussinesq approximation [61]; therefore, a body force term appears in the momentum equation.

For power-law fluid, the rheological behavior is expressed as [62]:

$$\boldsymbol{\tau} = K \dot{\boldsymbol{\gamma}}^n \quad (4)$$

where K is the consistency coefficient, $\dot{\boldsymbol{\gamma}}$ is the shear rate tensor, n is the power-law index reflecting the deviation degree from Newtonian fluid. Based on the value of power-law index, the power-law fluid can be classified into three categories. The case $n < 1$ corresponds to pseudoplastic or shear-thinning fluid, which is widely used in practice. The case $n > 1$ corresponds to dilatant or shear-thickening fluid. Obviously, the fluid exhibits Newtonian behavior for $n = 1$. The shear rate tensor $\dot{\boldsymbol{\gamma}}$ is related to strain rate tensor $\boldsymbol{\varepsilon}$ by:

$$\dot{\boldsymbol{\gamma}} = 2\boldsymbol{\varepsilon} = \nabla \mathbf{u} + (\nabla \mathbf{u})^T \quad (5)$$

where $\nabla \mathbf{u}$ is the velocity gradient, the superscript T represents the transposition operator. Here the magnitude of local shear rate is related to the second invariant of the symmetric strain rate tensor as

$$|\dot{\boldsymbol{\gamma}}| = \sqrt{2(\boldsymbol{\varepsilon} : \boldsymbol{\varepsilon})} \quad (6)$$

Thus, the Eq. (4) can be rewritten as

$$\boldsymbol{\tau} = K \dot{\boldsymbol{\gamma}}^n = \mu(|\dot{\boldsymbol{\gamma}}|) \dot{\boldsymbol{\gamma}} \quad (7)$$

where $\mu(|\dot{\boldsymbol{\gamma}}|) = K \dot{\boldsymbol{\gamma}}^{n-1}$ is treated as an apparent or effective viscosity.

2.2. Thermal LBM model for power-law fluid on orthogonal grids

In this section, the thermal LBM for power-law fluids on uniform Cartesian grids is briefly introduced. For simplicity, the two-dimensional nine-velocity (D2Q9) lattice Boltzmann model

is used in the LBM simulation for both hydrodynamic and thermodynamic fields in this paper. The discrete velocity $\mathbf{c}_i$ in the i th direction is given as

$$\mathbf{c}_i = \begin{cases} (0, 0) & i = 0 \\ c(\cos[(i-1)\pi/2], \sin[(i-1)\pi/2]) & i = 1-4 \\ \sqrt{2}c(\cos[(2i-1)\pi/4], \sin[(2i-1)\pi/4]) & i = 5-8 \end{cases} \quad (8)$$

where $c = \delta x/\delta t$ is lattice speed with δx and δt being the grid spacing and time step, respectively. Usually, c is set to be 1, which leads to $\delta x = \delta t$. Two sets of particle distribution functions, f_i and g_i , are introduced to characterize the velocity and temperature fields, respectively. The evolution equations are given as

$$f_i(\mathbf{x} + \mathbf{c}_i \delta t, t + \delta t) - f_i(\mathbf{x}, t) = -\frac{1}{\tau_f} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)] + \delta t F_i \quad (9)$$

$$g_i(\mathbf{x} + \mathbf{c}_i \delta t, t + \delta t) - g_i(\mathbf{x}, t) = -\frac{1}{\tau_T} [g_i(\mathbf{x}, t) - g_i^{eq}(\mathbf{x}, t)] \quad (10)$$

where $f_i^{eq}(\mathbf{x}, t)$ and $g_i^{eq}(\mathbf{x}, t)$ are the equilibrium particle distribution functions of density and temperature at position $\mathbf{x}$ and time t , respectively, and they can be expressed as

$$f_i^{eq} = \rho \omega_i \left[1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right] \quad (11)$$

$$g_i^{eq} = T \omega_i \left[1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right] \quad (12)$$

where ω_i is the weighting coefficient, c_s is the lattice velocity of sound. In the D2Q9 model, $\omega_i = 4/9$ for $i = 0$, $\omega_i = 1/9$ for $i = 1-4$, $\omega_i = 1/36$ for $i = 5-8$, and $c_s = \sqrt{3}/3$. τ_f and τ_T are the dimensionless relaxation times of density and temperature, respectively. The force term F_i is introduced to couple the velocity field and temperature field in the case of natural convection. Based on the coupled lattice BGK model [60], it is given as

$$F_i = \left(1 - \frac{1}{2\tau_f} \right) \omega_i \left[\frac{\mathbf{c}_i - \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})}{c_s^4} \mathbf{c}_i \right] \cdot \mathbf{G} \quad (13)$$

In the limit of Boussinesq approximation, $\mathbf{G} = -\beta(T - T_0)\mathbf{g}$. It should be emphasized that when forced convection is considered, the force term is neglected since it is one-way interaction problem, and only the velocity field can influence the temperature field.

The macroscopic physical parameters, density ρ , velocity $\mathbf{u}$, temperature T , and pressure p can be evaluated from

$$\rho = \sum_i f_i \quad (14)$$

$$\mathbf{u} = \frac{1}{\rho} \sum_i f_i \mathbf{c}_i \quad (15)$$

$$T = \sum_i g_i \quad (16)$$

$$p = \rho c_s^2 \quad (17)$$

It is well-known that for Newtonian fluid, the dimensionless relaxation time τ_f in Eq. (9) is related to the constant kinematic viscosity by $\nu = c_s^2(\tau_f - 0.5)\delta t$. Thus, relaxation time τ_f is constant for all lattice grids. However, as for power-law fluids considered in this work, the viscosity varies with the local shear rate. Therefore, it is necessary to determine the effective viscosity on the basis of constitutive equation, so as to calculate the local dimensionless relaxation time τ_f at each lattice grid. As shown in Eqs. (5)–(7), the effective viscosity is dependent on shear rate, which can be

obtained by calculating local derivatives of the velocity via finite-difference approximation. Alternatively, as stated previously, the strain rate tensor can be calculated locally at each node based on the particle distribution functions as [44,63]

$$\varepsilon_{\alpha\beta} = -\frac{1}{2\rho c_s^2 \tau_f \delta t} \sum_{i=0}^8 \mathbf{c}_{i\alpha} \mathbf{c}_{i\beta} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)] \quad (18)$$

Note that for the sake of simplicity, the relaxation time at last time step is used in Eq. (18). Obviously, the local calculation of the shear rate makes the LBM particularly advantageous if it is implemented in parallel. During the simulation, the effective viscosity and the relaxation time are renewed at each time step. Accordingly, the Eq. (9) is rewritten as

$$f_i(\mathbf{x} + \mathbf{c}_i \delta t, t + \delta t) - f_i(\mathbf{x}, t) = -\frac{1}{\tau_f} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)] + \delta t F_i \quad (19)$$

Moreover, it is noted that with regard to other types of non-Newtonian fluid, such as Bingham plastic fluid and Casson fluid, the evolution equations can be easily derived by determining the corresponding local relaxation time based on its constitutive equation in the similar manner.

The dimensionless relaxation time τ_T in Eq. (10) can be calculated through the thermal diffusivity D by

$$D = c_s^2 \left(\tau_T - \frac{1}{2} \right) \delta t \quad (20)$$

In the LBM, the evolution equations can be divided into two steps, namely, the collision and streaming steps. In the collision step, we have

$$f_i^*(\mathbf{x}, t) = -\frac{1}{\tau_f} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)] + \delta t F_i \quad (21a)$$

$$g_i^*(\mathbf{x}, t) = -\frac{1}{\tau_T} [g_i(\mathbf{x}, t) - g_i^{eq}(\mathbf{x}, t)] \quad (21b)$$

where $f_i^*(\mathbf{x}, t)$ and $g_i^*(\mathbf{x}, t)$ are the post-collision particle distribution functions of density and temperature, respectively.

In the streaming step, we have

$$f_i(\mathbf{x}, t + \delta t) = f_i^*(\mathbf{x} - \mathbf{c}_i \delta t, t) \quad (22a)$$

$$g_i(\mathbf{x}, t + \delta t) = g_i^*(\mathbf{x} - \mathbf{c}_i \delta t, t) \quad (22b)$$

2.3. Thermal LBM model for power-law fluid on non-orthogonal grids

Based on the GILBM, the above-mentioned thermal LBM model for power-law fluid is extended on non-orthogonal grids. The physical and computational planes are denoted by $\mathbf{x} = (x_1, x_2) = (x, y)$ and $\xi = (\xi_1, \xi_2) = (\xi, \eta)$, respectively. As stated in Eq. (21), only the local particle distribution functions are involved in the collision step, thus the transformation of the collision equations from the physical plane into the computational plane is easily achieved by replacing $\mathbf{x}$ with ξ . Then, the collision equations are obtained as

$$f_i^*(\xi, t) = -\frac{1}{\tau_f} [f_i(\xi, t) - f_i^{eq}(\xi, t)] + \delta t F_i \quad (23a)$$

$$g_i^*(\xi, t) = -\frac{1}{\tau_T} [g_i(\xi, t) - g_i^{eq}(\xi, t)] \quad (23b)$$

Fig. 1 gives the discrete velocity vector $\mathbf{c}_1 = c(1, 0)$ on the physical and computational planes, respectively. It is observed that the discrete velocity vectors are constant on the physical plane; however, they vary with the lattice position on the computational

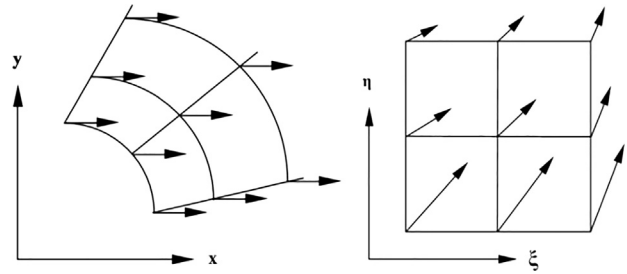


Fig. 1. Illustration of discrete velocity vector $\mathbf{c}_1 = c(1, 0)$ on physical and computational planes.

plane. Here, the discrete velocity on the computational plane $\tilde{\mathbf{c}}_i = (\tilde{c}_{i,1}, \tilde{c}_{i,2})$ is called contravariant velocity, and its components are defined as

$$\tilde{c}_{i,\alpha} = c_{i,\beta} \frac{\partial \xi_\alpha}{\partial x_\beta} \quad (\alpha, \beta = 1, 2) \quad (24)$$

The derivatives in Eq. (24) are calculated by

$$\begin{bmatrix} \frac{\partial \xi_1}{\partial x_1} & \frac{\partial \xi_1}{\partial x_2} \\ \frac{\partial \xi_2}{\partial x_1} & \frac{\partial \xi_2}{\partial x_2} \end{bmatrix} = \frac{1}{J} \begin{bmatrix} \frac{\partial x_2}{\partial \xi_2} & -\frac{\partial x_1}{\partial \xi_2} \\ -\frac{\partial x_2}{\partial \xi_1} & \frac{\partial x_1}{\partial \xi_1} \end{bmatrix} \quad (25)$$

where J is the Jacobian transformation, and it is defined as

$$J = \frac{\partial x_1}{\partial \xi_1} \frac{\partial x_2}{\partial \xi_2} - \frac{\partial x_1}{\partial \xi_2} \frac{\partial x_2}{\partial \xi_1} \quad (26)$$

In order to maintain the accuracy in space, second-order central-difference scheme is adopted to calculate the derivatives on the right hand side of Eqs. (25) and (26).

In the streaming step, the particle streaming distance in the i th direction is determined by integrating the contravariant velocity over a time step as

$$\Delta \xi_{up,i} = \int_0^{\delta t} d\xi_i = \int_0^{\delta t} \tilde{\mathbf{c}}_i dt \quad (27)$$

Then, the streaming equations become

$$f_i(\xi, t + \delta t) = f_i^*(\xi - \Delta \xi_{up,i}, t) \quad (28a)$$

$$g_i(\xi, t + \delta t) = g_i^*(\xi - \Delta \xi_{up,i}, t) \quad (28b)$$

The two-step Runge-Kutta method is used to determine the streaming distance $\Delta \xi_{up,i}$ as follows [55]

$$1^{\text{st}}\text{-step: } \Delta \xi_{up,i}^{(1)} = \frac{1}{2} \delta t \tilde{\mathbf{c}}_i(\xi) \quad (29a)$$

$$2^{\text{nd}}\text{-step: } \Delta \xi_{up,i} = \delta t \tilde{\mathbf{c}}_i(\xi - \Delta \xi_{up,i}^{(1)}) + O(\delta t^3) \quad (29b)$$

Note that $\Delta \xi_{up,i}$ is dependent on the contravariant velocity, so the positions of $\xi - \Delta \xi_{up,i}$ and $\xi - \Delta \xi_{up,i}^{(1)}$ may not exactly be located at the lattice nodes. Consequently, an interpolation procedure is required. As suggested by Imamura et al. [55], the second-order upwind quadratic interpolation is adopted to suppress numerical dissipation. The interpolation functions are given as

$$f_i(\xi, t + \delta t) = f_i^*(\xi - \Delta \xi_{up,i}, t) = \sum_{m=0}^2 \sum_{l=0}^2 a_{i,m,\xi} a_{i,l,\eta} f_i^*(\xi_{i+m-md}, \eta_{j+l-nd}, t + \delta t) \quad (30a)$$

$$g_i(\xi, t + \delta t) = g_i^*(\xi - \Delta \xi_{up,i}, t) = \sum_{m=0}^2 \sum_{l=0}^2 a_{i,m,\xi} a_{i,l,\eta} g_i^*(\xi_{i+m-md}, \eta_{j+l-nd}, t + \delta t) \quad (30b)$$

where $md = \text{sgn}(\Delta \xi_{up,i,x})$ and $nd = \text{sgn}(\Delta \xi_{up,i,y})$ are the coefficients to determine the lattice points for interpolation. As for the internal lattice nodes, the interpolation coefficients $a_{i,m,\xi}$ and $a_{i,l,\eta}$ are given as

$$\begin{aligned} a_{i,0,\alpha} &= \frac{1}{2} (|\Delta \xi_{up,i,\alpha}| - 1) (|\Delta \xi_{up,i,\alpha}| - 2) \\ a_{i,1,\alpha} &= -|\Delta \xi_{up,i,\alpha}| (|\Delta \xi_{up,i,\alpha}| - 2) \\ a_{i,2,\alpha} &= \frac{1}{2} |\Delta \xi_{up,i,\alpha}| (|\Delta \xi_{up,i,\alpha}| - 1) \end{aligned} \quad (31)$$

As for the neighbor-boundary nodes, the interpolation coefficients $a_{i,m,\xi}$ and $a_{i,l,\eta}$ are calculated by the central difference interpolation, which are given as

$$\begin{aligned} a_{i,0,\alpha} &= \frac{1}{2} \Delta \xi_{up,i,\alpha} (\Delta \xi_{up,i,\alpha} - 1) \\ a_{i,1,\alpha} &= 1 - \Delta \xi_{up,i,\alpha}^2 \\ a_{i,2,\alpha} &= \frac{1}{2} \Delta \xi_{up,i,\alpha} (\Delta \xi_{up,i,\alpha} + 1) \end{aligned} \quad (32)$$

The time step δt is determined by the Courant-Friedrichs-Lewy (CFL) to guarantee the numerical stability [64], and it is evaluated as

$$\delta t = \lambda \min_{i,\alpha,\xi} \left| \frac{1}{c_{i,\alpha}|\xi|} \right| \quad (33)$$

where λ is the CFL number ($0 < \lambda \leq 1$).

The basic numerical procedure of the proposed model can be outlined as follows:

1. Initialization: The values of density and velocity are first initialized based on the physical problem, and then the equilibrium distribution functions f_i^{eq} and g_i^{eq} are calculated based on Eqs. (11) and (12).
2. Calculation of the dimensionless relaxation time τ_f : The dimensionless relaxation time is calculated by Eq. (20).
3. Calculation of the strain rate tensor ε : Since the particle distribution function $f_i(\xi, 0)$, and its equilibrium part f_i^{eq} are obtained, the strain rate tensor at each lattice node can be calculated based on Eq. (18).
4. Calculation of the effective viscosity $\mu(|\dot{\gamma}|)$: Based on Eqs. (5)–(7), the effective viscosity at each lattice node can be calculated to determine dimensionless relaxation time τ_f' via $\mu(|\dot{\gamma}|) = \rho c_s^2 (\tau_f' - 0.5) \delta t$.
5. Evolution: The evolutions of particle distribution function are performed based on Eqs. (23) and (30).
6. Calculation of macroscopic quantities: The density, velocity, temperature, and pressure are evaluated through Eqs. (14)–(17).
7. Go to step 3 if the specified precision is not satisfied.

3. Boundary conditions

Boundary implementation plays an important role in the LBM because it influences the accuracy and stability of the method. Since the governing equations have been transformed from the physical plane into the computational plane, the boundary condition should also be transformed. In current study, the non-equilibrium extrapolation scheme proposed by Guo et al. [60,65] is developed to deal with velocity and thermal boundary conditions as well as symmetry boundary on non-orthogonal grids.

The basic idea of the extrapolation method is that the distribution function is assumed to be composed of its equilibrium and non-equilibrium parts. The non-equilibrium term can be approximately obtained by the corresponding value at its the nearest

neighbor fluid node $\mathbf{x}_f$. The equilibrium part can approximately be calculated by replacing the unknown macroscopic variables with the values at $\mathbf{x}_f$. Consequently, the post-collision particle distribution functions of density and temperature at the boundary node $\mathbf{x}_b$ on the physical plane can be expressed as

Velocity boundary:

$$f_i^*(\mathbf{x}_b, t) = f_i^{eq}(\rho(\mathbf{x}_f), \mathbf{u}(\mathbf{x}_b), t) + \left(1 - \frac{1}{\tau_f'}\right) [f_i(\mathbf{x}_f, t) - f_i^{eq}(\mathbf{x}_f, t)] + \delta t F_i \quad (34a)$$

Constant temperature boundary:

$$g_i^*(\mathbf{x}_b, t) = g_i^{eq}(T(\mathbf{x}_b), \mathbf{u}(\mathbf{x}_b), t) + \left(1 - \frac{1}{\tau_T}\right) [g_i(\mathbf{x}_f, t) - g_i^{eq}(\mathbf{x}_f, t)] \quad (34b)$$

The transformation of boundary conditions is stated as follows:

First, through the Chapman-Enskog analysis for the lattice Boltzmann method, the non-equilibrium terms in Eq. (34) can be assumed to be $f_i^{(ne)} = K f_i^{(1)}$ and $g_i^{(ne)} = K g_i^{(1)}$ with K being a small parameter proportion to the Knudsen number, respectively. Therefore, on the computational plane, the second term on the right hand side of Eq. (34) can be directly obtained by replacing $\mathbf{x}$ with ξ .

Next, special treatment comes from the calculation of equilibrium part. For Dirichlet type velocity and thermal boundary conditions, considering the small fluctuation of pressure in incompressible flows, the first term on the right hand side of Eq. (34) can be determined by replacing $\mathbf{x}$ with ξ . The accuracy and reliability have been proved by our previous studies [66,67]. However, it is emphasized that as for the adiabatic boundary or symmetry boundary which contains the derivative term, the macroscopic variable at boundary node should be determined on the basis of condition transformation.

Taking the adiabatic boundary for example, the temperature at boundary node in the equilibrium $g_i^{eq}(T(\mathbf{x}_b), \mathbf{u}(\mathbf{x}_b), t)$ is unknown. Instead, the normal derivative at the boundary on the physical plane can be expressed as

$$\frac{\partial T}{\partial n} = 0 \quad (35)$$

On the computational plane, it can be derived as

$$\frac{\partial T}{\partial n^{(\xi)}} = \frac{\alpha \frac{\partial T}{\partial \xi} - \beta \frac{\partial T}{\partial \eta}}{J\sqrt{\alpha}} = 0 \quad (36a)$$

$$\frac{\partial T}{\partial n^{(\eta)}} = \frac{\gamma \frac{\partial T}{\partial \eta} - \beta \frac{\partial T}{\partial \xi}}{J\sqrt{\gamma}} = 0 \quad (36b)$$

where $\alpha = \left(\frac{\partial x}{\partial \eta}\right)^2 + \left(\frac{\partial y}{\partial \eta}\right)^2$, $\beta = \frac{\partial x}{\partial \xi} \frac{\partial x}{\partial \eta} + \frac{\partial y}{\partial \xi} \frac{\partial y}{\partial \eta}$, and $\gamma = \left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2$. Consequently, the unknown temperature at boundary node can be determined by solving Eq. (36). The symmetry boundary with $\partial u / \partial n = 0$ can be derived in the similar manner.

4. Fluid flow and heat transfer simulations

In this section, both forced convection and natural convection heat-transfer of power-law fluids are simulated to validate the proposed thermal LBM model on non-uniform grids. The target cases include forced convection over a stationary heated circular cylinder, and natural convection in cavities and an annulus between concentric circular cylinders, respectively. For all the conditions

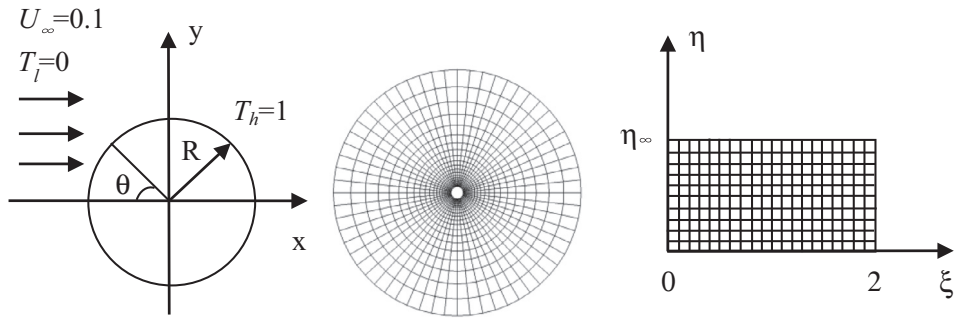


Fig. 2. Schematic diagram for physical model, grid generations.

spanned, the flow is assumed to be laminar and steady. The solution is assumed to be converged when the relative errors between the new and old values of temperature and velocity fields are less than 10^{-8} , respectively. Specially, the relative error of velocity is quantified using L_2 norm, which is defined as

Error

$$= \frac{\sqrt{\sum_{\xi,\eta} \{ [u_{\xi}(\xi, \eta, t + \delta t) - u_{\xi}(\xi, \eta, t)]^2 + [u_{\eta}(\xi, \eta, t + \delta t) - u_{\eta}(\xi, \eta, t)]^2 \}}}{\sqrt{\sum_{\xi,\eta} [u_{\xi}(\xi, \eta, t + \delta t)]^2 + [u_{\eta}(\xi, \eta, t + \delta t)]^2}} \quad (37)$$

where u_{ξ} and u_{η} are the velocity components.

4.1. Forced convection over a stationary heated circular cylinder

The flow of power-law fluids over a circular cylinder has been thoroughly studied by the authors [29,68–70]. In the current work, the developed LBM model on non-uniform grids is applied to study

the steady flow of power-law fluids over a stationary heated circular cylinder. The obtained drag coefficient C_D , recirculation length L_w (normalized by the circular cylinder diameter $2R$), average Nusselt number Nu , and local Nusselt number $Nu(\theta)$ are compared with the previous numerical studies. Sivakumar et al. [71] pointed out that for the power-law indices ($0.5 \leq n \leq 1.4$), the critical Reynolds number is around 40 when steady symmetric flow transmits to unsteady laminar vortex shedding regime, thus the flow condition in current study is restricted to $Re \leq 40$.

The grid generations on the physical plane and computational plane are illustrated in Fig. 2. On the physical plane, r and θ are the polar coordinates. On the computational plane, the grids are

Table 1
Comparison of recirculation lengths in different power-law indices.

Reference	$n = 0.7$	$n = 1.0$	$n = 1.3$
Present work	1.864	2.272	2.811
Amiri et al. [46]	1.945	2.342	2.874
He et al. [53]	–	2.245	–

Table 2
Comparison of drag coefficients in different power-law indices.

References	$n = 0.6$	$n = 1.0$	$n = 1.4$
Present work	1.316	1.498	1.613
Soares et al. [69]	1.372	1.503	1.633

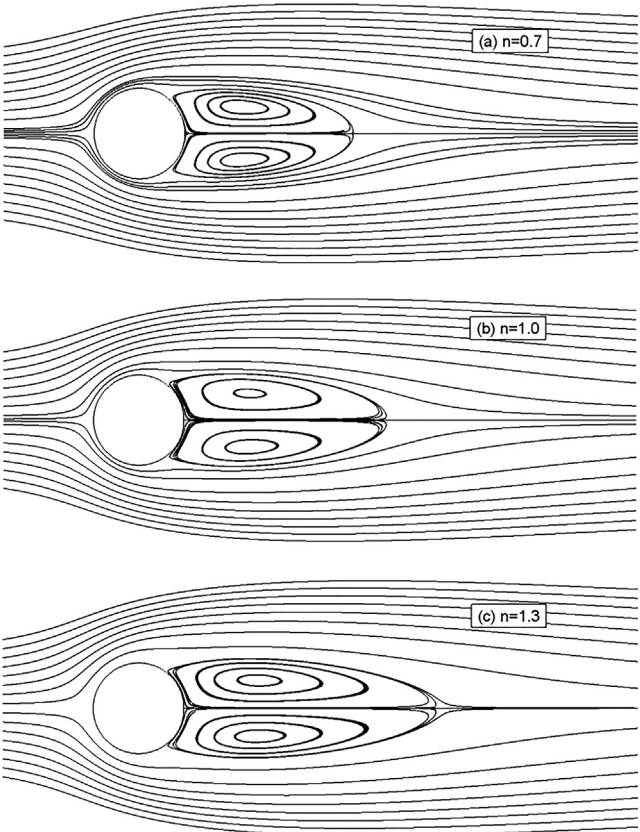


Fig. 3. Streamlines for different power-law indices (a) $n = 0.7$, (b) $n = 1.0$, and (c) $n = 1.3$.

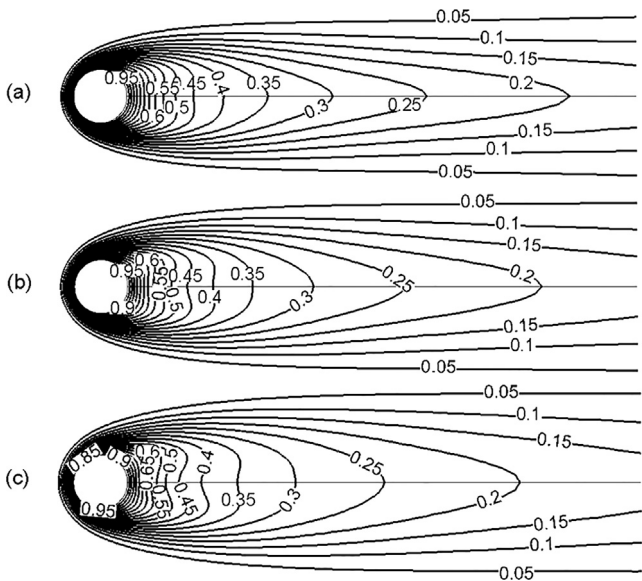


Fig. 4. Isotherms for different power-law indices (a) $n = 0.6$, (b) $n = 1.0$, and (c) $n = 1.4$.

uniform, and the coordinates ξ and η are related to r and θ by $r_j = Re^{\pi\eta/l}$ and $\theta_i = \pi\xi_i$, respectively, where R is the cylinder radius. In current study, the computational domain is divided into 361×271 grids. Fluid with constant temperature $T_l = 0$ flows over a stationary circular cylinder at the streaming velocity $U_\infty = 0.1$. The temperature of cylinder surface is kept at a constant high temperature of $T_h = 1$. The non-slip boundary condition is held at the

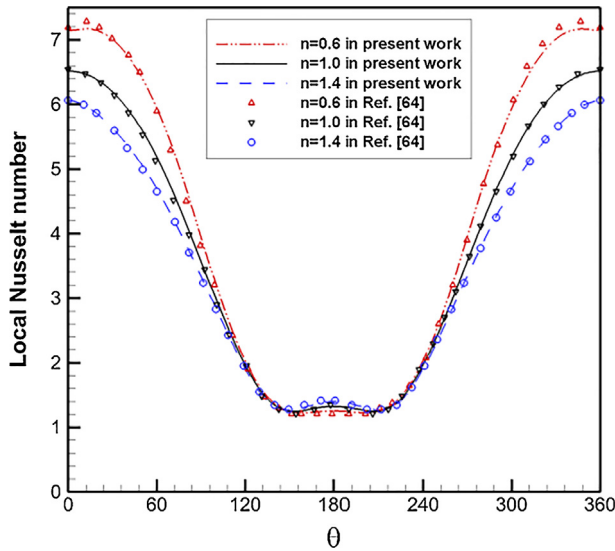


Fig. 5. Comparison of local Nusselt number distributions on the cylinder surface for different power-law indices between present work and Ref. [64].

Table 3

Comparison with previous studies of average Nusselt numbers.

References	$n = 0.6$	$n = 1.0$	$n = 1.4$
Present work	4.0668	3.6833	3.4295
Amiri et al. [46]	4.0636	3.7097	3.4614
Patnana et al. [68]	4.0775	3.6755	3.4216
Bharti et al. [29]	4.0545	3.6534	3.4004

cylinder surface. The non-equilibrium extrapolation boundary on non-orthogonal grids previously stated is adopted. The Reynolds number and Prandtl number are fixed as $Re = 40$ and $Pr = 1.0$, respectively. For power-law fluid, they are defined as [46]

$$Re = \frac{\rho(2R)^n U_\infty^{2-n}}{K}, \quad Pr = \frac{Kc_p}{k} \left(\frac{U_\infty}{2R} \right)^{n-1} \quad (38)$$

The flow and heat transfer are simulated at $n = 0.6, 0.7, 1.0, 1.3$, and 1.4 , respectively.

The drag coefficient is defined as

$$C_D = \frac{\mathbf{F} \cdot \mathbf{x}}{\rho U_\infty^2 R} \quad (39)$$

where $\mathbf{F}$ is the total force acted on the solid by fluid. It can be obtained by integrating the stress tensor around the surface of the cylinder as $\mathbf{F} = \int \sigma \cdot \mathbf{n}_s d\mathbf{l}$ [53]. The stress tensor can be calculated by

$$\begin{aligned} \sigma_{\alpha\beta} &= -p\delta_{\alpha\beta} + \tau_{\alpha\beta} \\ &= -p\delta_{\alpha\beta} + 2\rho\nu e_{\alpha\beta} \end{aligned} \quad (40)$$

Nusselt number and average Nusselt number are key parameters for the examination of the heat transfer rate from the heated surface. Here, the local Nusselt number is defined as [72]

$$Nu(\theta) = -\frac{L}{T_h - T_l} \left(\frac{\partial T}{\partial n_s} \right) \quad (41)$$

where n_s is the normal direction to the surface, L is the characteristic length with $L = 2R$. In the calculation of $\partial T / \partial n_s$, the derivative on the computational plane should be used. The average Nusselt number can be obtained by averaging the local values over the cylinder surface as

$$Nu = \frac{1}{2\pi} \int_0^{2\pi} Nu(\theta) d\theta \quad (42)$$

When steady states are achieved, the streamlines for different power-law indices are shown in Fig. 3. It is clearly observed that a pair of symmetric eddies are formed behind the cylinder. The recirculation length increases with power-law index. The results show fair agreement with those in Refs. [46,53], as stated in Table 1. The drag coefficients for various power-law indices are compared

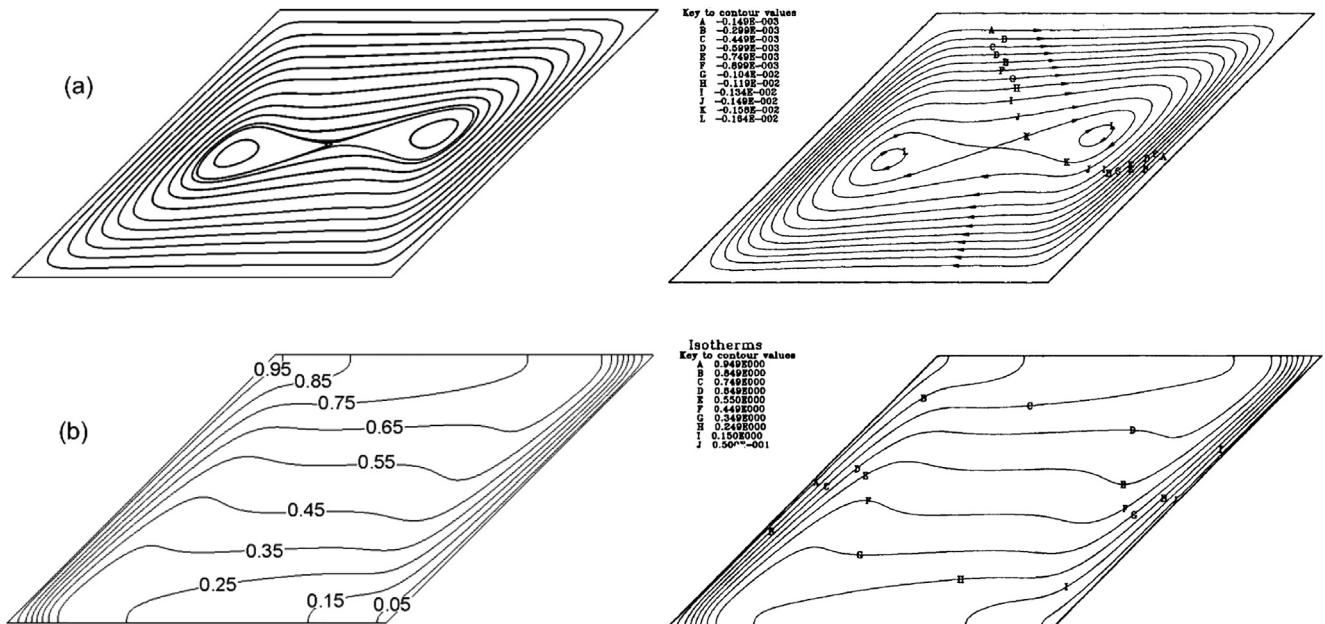


Fig. 6. Comparison of (a) streamlines and (b) isotherms with previous results (left: present results; right: results by Demirdžić et al. [73]).

with the values reported by Soares et al. [69] in Table 2. It is found out that the drag coefficient increases as the fluid behavior changes from shear-thinning fluid to Newtonian fluid and finally to shear-thickening fluid.

Fig. 4 shows isotherms in the vicinity of the cylinder for various power-law indices. When the power-law index is fixed, a large temperature gradient appears in the front of the cylinder as the

isotherms cluster heavily there, and this implies that the heat transfer rate is much larger than that in other regions, so the local Nusselt numbers on the front surfaces are higher, as shown in Fig. 5. The clustering of isotherms at the back of the cylinder is enhanced with increase of power-law index. Consequently, the local Nusselt number at rear stagnation point grows higher for shear-thickening fluid. The simulation results agree well with the data in the literature. The comparison with previous studies of average Nusselt numbers for various power-law indices are illustrated in Table 3. As expected, the average Nusselt number grows as the power-law index decreases, which indicates that the heat transfer is enhanced by the shear-thinning fluid.

4.2. Natural convection in cavies

Firstly, natural convection of Newtonian fluid ($n = 1.0$) in an inclined cavity is considered. Following the study of Demirdžić et al. [73], the domain is a rhombus with angle of $\phi = 45^\circ$. The sides of the cavity are of length $L = 1$, and the grid resolutions are 321×321 . The upper and lower horizontal walls of the enclosure are adiabatic, whereas the left and right walls are maintained at constant high and low temperatures of $T_h = 1$ and $T_c = 0$, respectively. Non-slip boundary conditions are applied for all the walls. The previously mentioned boundary implementation is used. In this simulation, the Prandtl number is 10, and the Rayleigh number is 10^6 .

The previous work of Demirdžić et al. [73] is taken as a benchmark to validate current method. The streamline and isotherm are compared with their results in Fig. 6. As expected, one free stagna-

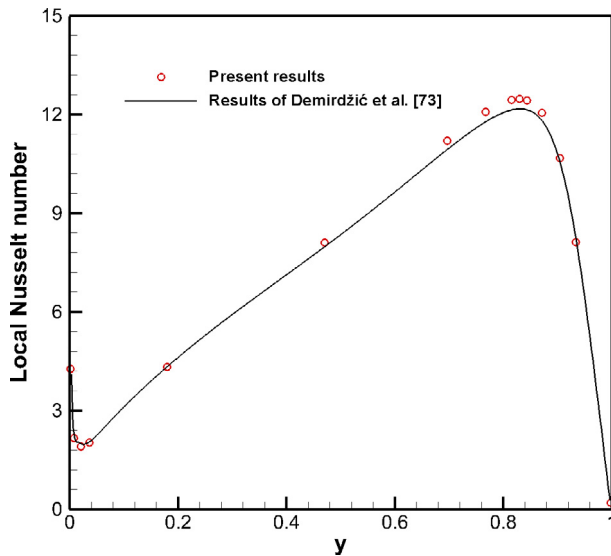


Fig. 7. Local Nusselt number distribution along the cold surface.

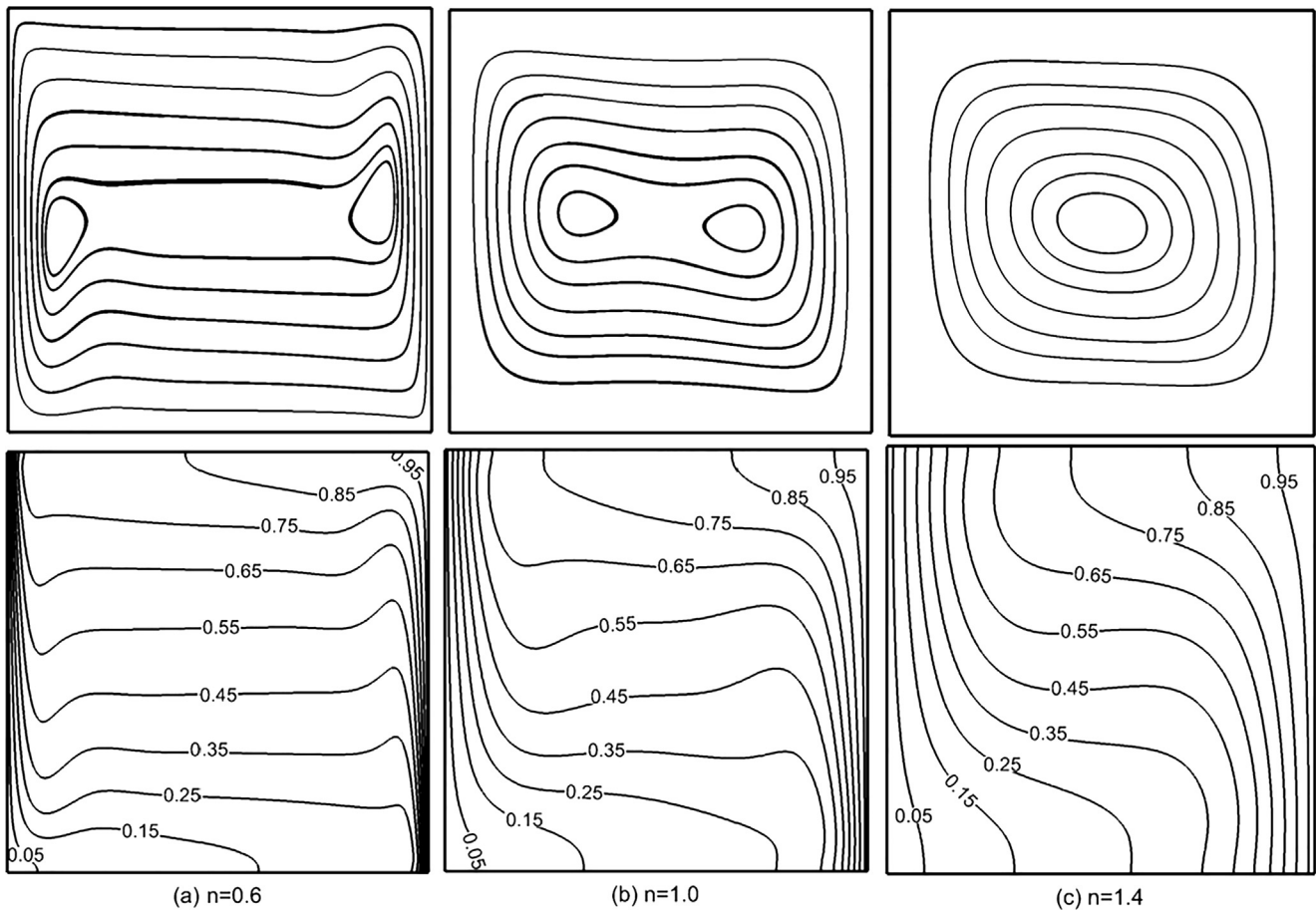


Fig. 8. Streamlines (upper) and isotherms (lower) for different values of power-law index.

tion point is located at the center of the cavity. The fluid is circulated due to the buoyancy determined by temperature difference. As shown in Fig. 6(b), the temperature gradients near the isothermal wall are steep, and the isotherm contours become flat at the central region as a consequence of the convection process. Particularly, it is observed that the isotherms are orthogonal to the upper and lower boundaries, which illustrates the correctness of the adiabatic boundary condition. In order to make a quantitative comparison, the local Nusselt numbers along the cold surface are shown in Fig. 7. It can be seen that our results agree well with benchmark solutions.

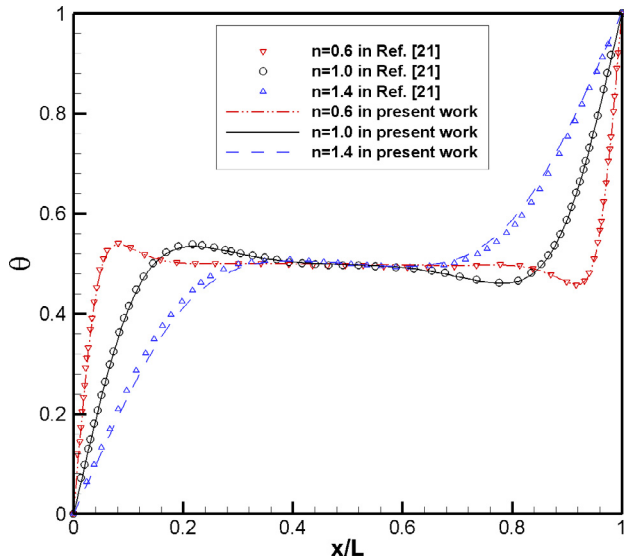


Fig. 9. Comparison of dimensionless temperature θ along horizontal mid-line between present work and Ref. [21].

Table 4
Comparison with previous studies of average Nusselt numbers over the cold wall.

References	$n = 0.6$	$n = 1.0$	$n = 1.4$
Present work	14.2918	4.6984	2.2838
Turan et al. [21]	14.3939	4.7258	2.2895
Matin et al. [37]	13.0672	4.6931	2.2836

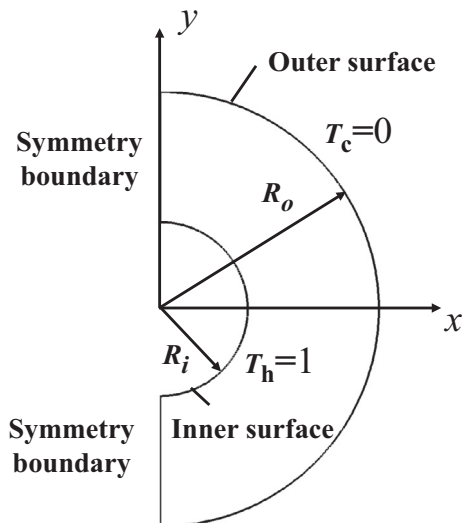


Fig. 10. Schematic of the flow configuration.

Then, natural convection of power-law fluids in a square cavity for $\phi = 90^\circ$ is concerned to study the effect of rheological property. Currently, the Prandtl number and Rayleigh number are defined as [21],

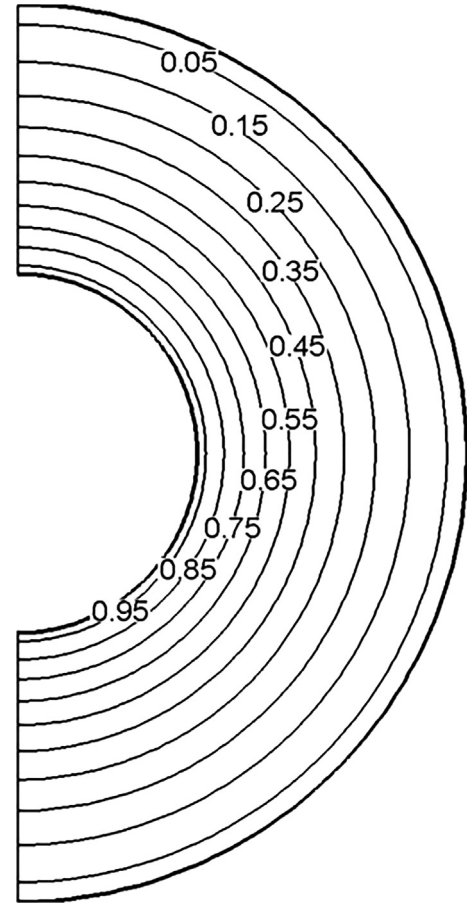


Fig. 11. Isotherms of $n = 1.0$ for pure conduction.

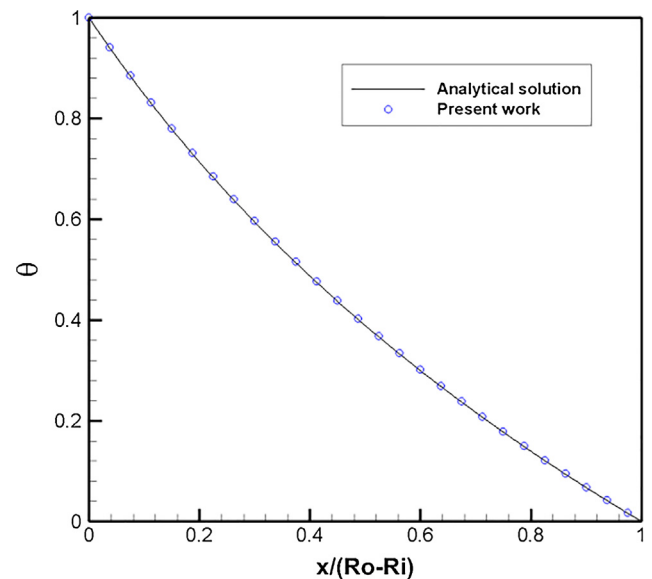


Fig. 12. Dimensionless temperature distribution along horizontal radial line for pure conduction.

$$Pr = \frac{KL^{2-2n}}{\rho D^{2-n}} \quad Ra = \frac{g\beta\Delta TL^{2n+1}}{D^n(K/\rho)} \quad (43)$$

Currently, the simulation conditions in the study of Turan et al. [21] are selected. Specifically, Prandtl number at $Pr = 10^2$ and Rayleigh at $Ra = 10^5$ are applied for the power-law indices being as $n = 0.6, 1.0$, and 1.4 , respectively. The boundary conditions are the same as those for the inclined cavity expect that the cavity is heated to a high temperature of $T_h = 1$ at the right wall and the temperature $T_c = 0$ is maintained at the left wall.

The streamlines and isotherms for $n = 0.6, 1.0$, and 1.4 at $Pr = 10^2$ and $Ra = 10^5$ are presented in Fig. 8. Generally, the fluid flows upward along the hot wall and then travels along the top insulated wall. It descends along the left cold wall, and then moves along the bottom insulated wall until it reaches the hot wall again. For given Prandtl and Rayleigh numbers, compared with the Newtonian fluid, the streamline pattern of shear-thickening fluid shows poor circulation due to the weak buoyancy induced flow current, and only one vortex is formed at the center of cavity. On the contrary, the convection process is enhanced for shear-thinning fluid, two vortices move toward the wall due to the strengthening of convective transport. As for the isotherms, steep gradients are

observed near the left and right walls, and the isotherms gather heavily as the decreasing of power-law index. The distribution of dimensionless temperature $\theta = (T - T_c)/(T_h - T_c)$ along the horizontal mid-line is shown in Fig. 9. A good match of average Nusselt number over the cold wall is found for present work with Turan et al. [21] and Matin et al. [37], as shown in Table 4.

4.3. Natural convection in an annulus between concentric circular cylinders

The proposed method is now applied to simulate natural convection of power-law fluids ($n = 0.6, 1.0$, and 1.4) in an annulus between two concentric horizontal cylinders of radii R_i and R_o . In order to reduce computational cost, the computational domain is taken as half of the physical domain due to symmetry. Fig. 10 shows the physical model with appropriate boundary conditions. The temperatures of outer and inner surfaces are kept at uniform $T_c = 0$ and $T_h = 1$, respectively. The Prandtl and Rayleigh numbers enclosed in the annular space are set to be 10^2 and 10^4 , respectively. The radii ratio $RR = R_o/R_i$ is 2.5 . The results are obtained from simulations on 401×801 grid resolution for $n = 0.6$ and on 161×321 grid resolution for $n = 1.0$ and 1.4 .

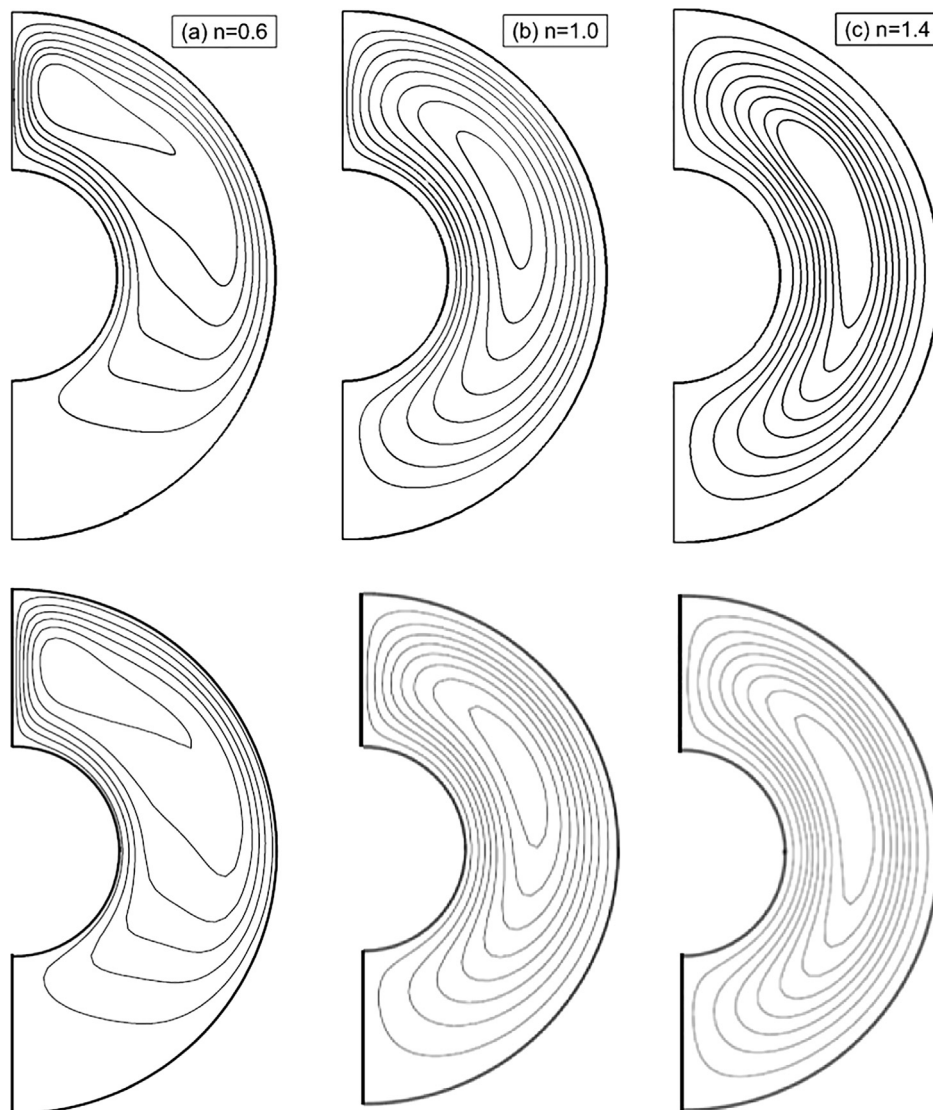


Fig. 13. Comparison of streamlines for (a) $n = 0.6$, (b) $n = 1.0$, and (c) $n = 1.4$ with previous results (upper: present results; lower: results by Matin et al. [37]).

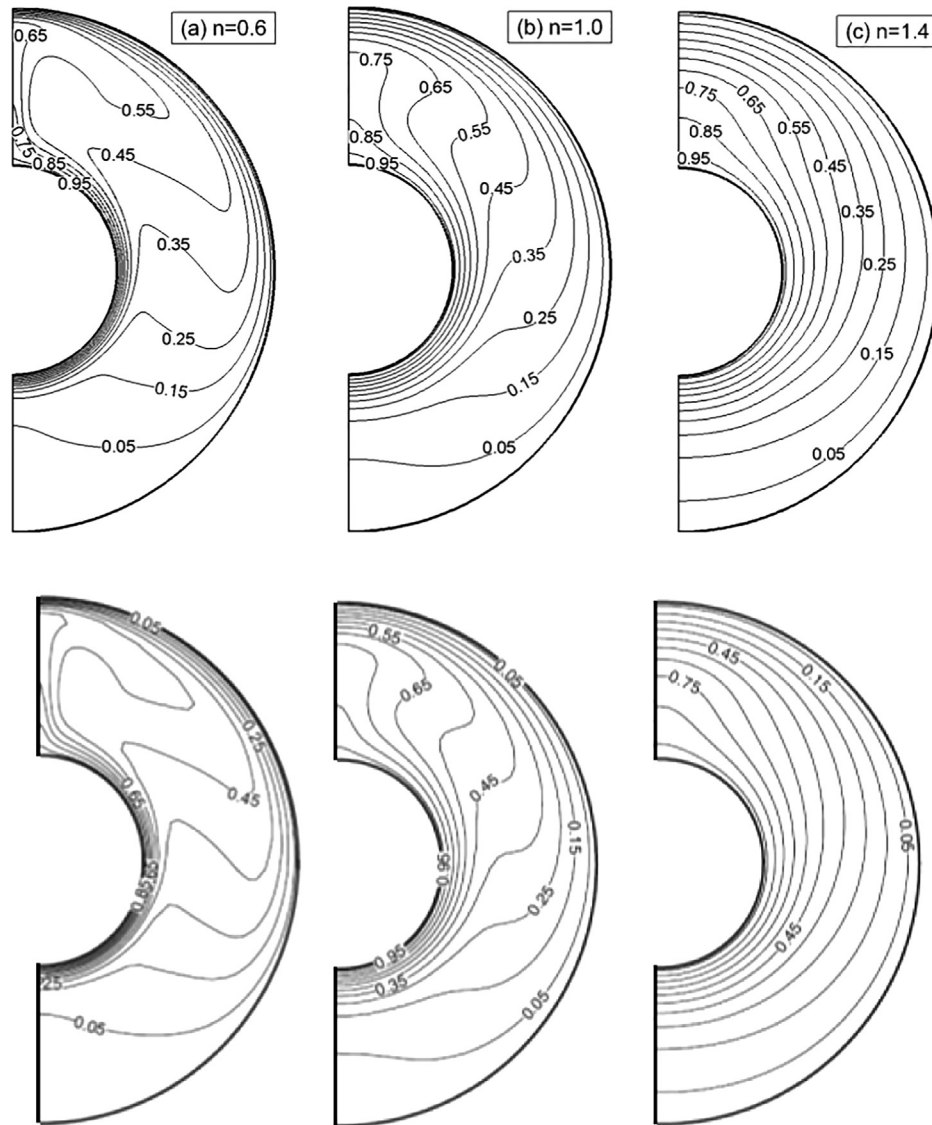


Fig. 14. Comparison of isotherms for (a) $n = 0.6$, (b) $n = 1.0$, and (c) $n = 1.4$ with previous results (upper: present results; lower: results by Matin et al. [37]).

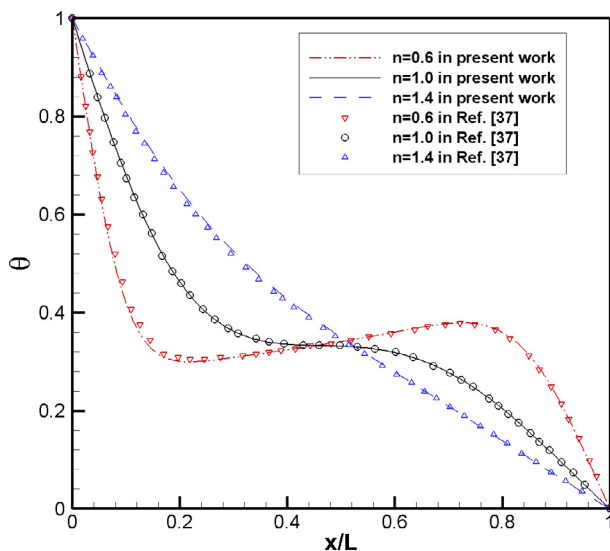


Fig. 15. Comparison of non-dimensional temperature θ along horizontal radial line between present work and Ref. [37].

Table 5

Comparison with previous study of average Nusselt numbers over the inner surface.

References	$n = 0.6$	$n = 1.0$	$n = 1.4$
Present work	6.91	3.26	1.99
Matin et al. [37]	7.17	3.31	2.01

For reference and validation purposes, the pure conduction problem for Newtonian fluid is simulated first. As shown in Fig. 11, the regular pattern in the isotherms can be found, being parallel in the enclosure, as expected. The dimensionless temperature θ along the horizontal radial line ($y = 0$) is compared with the analytical solution given as $T = T_h + \frac{T_c - T_h}{\ln(R_o/R_i)} \ln(x/R_i)$ in Fig. 12. The numerical results are in an excellent agreement with the analytical solution. Here, the distance from the inner surface to outer surface is normalized by the annulus spacing as $\Xi = x/(R_o - R_i)$.

The streamlines and isotherms for different power-law indices at specified Prandtl and Rayleigh numbers are shown in Figs. 13 and 14. It can be observed that the cold fluid is heated by the inner surface and it move upward along the hot surface. When the flow arrives at the top of enclosure, it descends along the cold outer

surface. With the decreasing of power-law index, the eye of the circulation moves upward and outward towards the cold wall due to the increasing convective strength. It is clear that the isotherms deviate significantly from the pure conduction case as the power-law index decreases, because the fluid movement and convection transport become strong. As for shear-thickening fluid, the pattern of isotherms is similar to the pure conduction case, except for the thermal boundary layer thickness at the bottom and top of the inner surface. All the aforementioned results agree well with those reported by Matin et al. [37].

The effect of power-law index on the dimensionless temperature θ along the horizontal radial line ($y = 0$) is depicted in Fig. 15. It is observed that as the power-law index decreases (from shear-thickening fluid to shear-thinning fluid), the temperature profiles become steep, which indicates that the dominant mechanism of heat transfer changes from conduction to convection. Table 5 compares average Nusselt numbers over the inner surface for $n = 0.6, 1.0$, and 1.4 with those of Matin et al. [37]. As expected, a decrease of power-law index causes an increase of average Nusselt number, which supports that non-Newtonian fluid is more efficient than Newtonian fluid for cooling or insulating purpose in engineering applications. The results are in good (with the maximum relative effort of 3.6% for $n = 0.6$) agreement with the benchmark solutions.

5. Conclusion remarks

In this paper, a thermal lattice Boltzmann model for power-law fluids on non-orthogonal grids is presented. The model can deal with thermal fluid flow in irregular domains. Another contribution of this paper is that it presents the boundary implementation of velocity and thermal boundary conditions in body-fitted coordinates. In particular, the non-equilibrium extrapolation scheme on uniform grids is extended in body-fitted coordinates by transforming the derivative from the physical plane to the computational plane.

The capability of present method is validated by applying it to simulate forced convection over a stationary heated circular cylinder and natural convection of power-law fluids in various enclosures, respectively. The streamline, isotherm, temperature profile, local Nusselt number, and average Nusselt number are examined. Numerical results show fair agreement with the available data in previous studies. The work suggests that current LBM scheme has a reliable potential for the simulation of thermal flow of power-law fluids in irregular geometries.

Conflict of interest

I would like to declare on behalf of my co-authors that no conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. We confirm that the contents of this manuscript is an original research which have not been published previously, and not under consideration for publication elsewhere, in whole or in part.

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